Pooria Shahbazi

Information Engineering Student | Embedded Systems & Software

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Profile

Information Engineering student at HAW Hamburg with hands-on experience in Embedded Linux, microcontroller programming (ESP32, C) and software development in Java and Python. Member of the Hawks Racing electronics and software team, working on printed circuit boards (PCBs) and control-unit firmware. Project experience ranging from FPGA design to a Raspberry Pi platform and ROS-based robotics. **Seeking a working-student position or internship in embedded systems, IT or software development.**

Education

B.Sc. Information Engineering

HAW Hamburg | 10/2023 – present (expected graduation: summer 2028)

- Project focus: embedded systems, digital design (FPGA/VHDL), robotics, software engineering

Computer Engineering (studies)

Leibniz University Hannover | 10/2021 – 09/2023

Studienkolleg – T-Course

Niedersächsisches Studienkolleg, Hannover | 09/2020 – 07/2021

- University preparatory course for engineering and natural sciences; passed the Feststellungsprüfung (university entrance assessment exam).

Dentistry (studies)

Haybusak University, Yerevan | 10/2015 – 09/2019

Professional Experience

Electronics and Software Team

Hawks Racing e.V. | 08/2026 – present

- Contribute to the design and implementation of electronics and printed circuit boards (PCBs) for the H22 car using Altium Designer.
- Program ESP32 control units in C.

Software Engineer

Coders Club Hamburg | 02/2026 – present | coders-club.dev

- Software development with a focus on Java backend.

Service Staff

Giovanni L, Hannover and Hamburg | 07/2021 – present

- Customer-facing work alongside full-time studies.

Warehouse Assistant

Amazon Logistics, Garbsen | 12/2020 – 06/2021

Deputy Financial Director

BetConstruct and subsidiary, Yerevan | 07/2019 – 12/2019

Finance Officer – Transaction Review

BetConstruct and subsidiary, Yerevan | 12/2018 – 06/2019

Customer Support Agent (Backoffice)

BetConstruct and subsidiary, Yerevan | 10/2018 – 12/2018

- Handled online enquiries and maintained the company database using BetConstruct Backoffice and Ladesk LiveAgent.

Graphic Designer and Content Creator

International Association of Group Exercises and Water Sports, Yerevan | 06/2016 – 12/2016

Projects

CyPi – Modular Raspberry Pi Cyberdeck

Personal project, ongoing | github.com/PooriaShahbazi/CyPi

- Built a portable Raspberry Pi 4B Linux platform integrating a 5-inch DSI display, LiPo power supply, MAX17048 battery gauge and audio.
- Developed Python prototypes for I^2C battery monitoring and GPIO rotary-encoder input; configured Linux autostart and prototyped a PySide6 launcher.
- Designed a modular expansion concept for interchangeable TI TM4C1294 and NXP MCX N236 microcontroller modules (magnetic connector and USB CDC protocol in planning).

RobotX – Team Leader and Programmer

HAW Hamburg | 04/2024

- Led a multidisciplinary team of five in robot development and programming with Python, ROS and OpenCV.

FPGA Traffic Light Controller

HAW Hamburg | 01/2025

- Designed and implemented a traffic light controller in VHDL (Vivado) and verified the digital logic in simulation.

LocalChatApp – Java Client-Server Application

Personal project | github.com/PooriaShahbazi/LocalChatApp

- Developed a multi-user group chat with TCP sockets, one thread per client and server-side message broadcasting.
- Built a Swing GUI with a background network listener and UI updates on the Event Dispatch Thread.

Interactive Portfolio Website

Personal project | github.com/PooriaShahbazi/Lebenslauf

- Built a German/English website with HTML, CSS and JavaScript, featuring interactive 3D navigation, DOM event handling and an animated Canvas background.

EzWg – Web Development

University business project, HAW Hamburg | 12/2024

- Built a website with HTML, CSS and JavaScript.

Additional Java Applications

Personal projects, source code on GitHub

- ChessGame, ArcadeGame, SpaceInvadersGame and SolarSystemSim.

Technical Skills

Embedded Systems: Embedded Systems, Embedded Linux, ESP32, Arduino, Raspberry Pi 4B, TI Tiva C Series TM4C1294, NXP FRDM-MCXN236, microcontrollers, GPIO, I2C, PCB design, Altium Designer

Digital Design: FPGA, VHDL, Xilinx Vivado

Programming Languages: C, C++, Java, Python, JavaScript, SQL, Bash

Software and Web: Object-oriented programming (OOP), multithreading, TCP sockets, REST APIs, React, HTML, CSS, Swing, PySide6, databases

Robotics and Analysis: ROS, OpenCV, MATLAB

Tools: Git, GitHub, Linux, Code Composer Studio, MCUXpresso, Android Studio, Unity, LaTeX, Microsoft Office

Languages

German: C1 | English: C1 | Persian: native | Turkish: B2 | Spanish: A1 | Armenian: A1